The Centurion® – AI-Powered Cybersecurity & Access Dashboard

DOCUMENTATION:

Orientation:

The Centurion® is a swift, web-based cyber-security dashboard trained to provide secure access control using complex facial recognition AI. It predicts age, gender, and ethnicity, logs unauthorized attempts, and can integrate with IoT devices and blockchain enviroments. It addresses real-world security constraints in workplaces, smart homes, and SMBs by combining AI, high-performance computing, and optional blockchain for static auditability.

Entails IV sections :

- ☐ Features and functionality
- ☐ Architecture and blueprint
- ☐ Data Flow Diagram[DFD]
- ☐ Project setup

```
aura@fedora:~$ sudo systemctl start aura.service
```

Centurion/

```
├── backend_cpp/
│   ├── src/
│   │   ├── main.cpp
│   │   ├── face_recognition.cpp
│   │   ├── demographics_estimator.cpp
│   │   ├── liveness_detection.cpp
│   │   ├── tor_integration.cpp
│   │   ├── logging.cpp
│   │   └── iot_simulation.cpp
│   ├── include/
│   │   ├── face_recognition.h
│   │   ├── demographics_estimator.h
│   │   ├── liveness_detection.h
│   │   ├── tor_integration.h
│   │   ├── logging.h
│   │   └── iot_simulation.h
│   ├── CMakeLists.txt
│   ├── models/
│   │   ├── facenet_embedding.onnx
│   │   ├── age_gender_model.onnx
│   │   └── ethnicity_model.onnx
│   ├── tor_config/
│   │   └── torrc
│   └── utils/
│       ├── image_utils.cpp
```

```
│       └── network_utils.cpp
│
├── frontend/
│   ├── public/
│   │   ├── index.html
│   │   ├── favicon.ico
│   │   └── assets/
│   │       ├── images/
│   │       │   └── centurion_logo.png
│   │       └── icons/
│   ├── src/
│   │   ├── components/
│   │   │   ├── LoginPage.jsx
│   │   │   ├── PermissionsPage.jsx
│   │   │   ├── Dashboard.jsx
│   │   │   ├── LiveFeed.jsx
│   │   │   └── IoTSimulation.jsx
│   │   ├── api/
│   │   │   └── backendAPI.js
│   │   ├── styles/
│   │   │   ├── main.css
│   │   │   └── animations.css
│   │   ├── utils/
│   │   │   └── webcam_utils.js
│   │   └── App.jsx
│   ├── package.json
```

```
│   └── tailwind.config.js
│
├── database/
│   ├── migrations/
│   ├── seeds/
│   ├── models/
│   │   ├── User.js
│   │   ├── LogEntry.js
│   │   └── FaceEmbedding.js
│   └── db_config.js
│
├── blockchain/
│   ├── contracts/
│   │   └── CenturionLog.sol
│   ├── scripts/
│   │   ├── deploy.js
│   │   └── logEntry.js
│   └── package.json
│
├── docs/
│   ├── TheCenturion_ProjectDocumentation.pdf
│   ├── roadmap.md
│   ├── architecture_diagram.png
│   └── tor_integration_guide.md
│
├── scripts/
```

```
│   ├── setup_env.sh
│   ├── start_backend.sh
│   ├── start_frontend.sh
│   └── run_tor_service.sh
│
├── tests/
│   ├── backend_tests/
│   │   ├── face_recognition_test.cpp
│   │   └── logging_test.cpp
│   ├── frontend_tests/
│   │   ├── loginPage.test.jsx
│   │   └── dashboard.test.jsx
│   └── integration_tests/
│       └── full_pipeline.test.js
│
├── .gitignore
├── README.md
└── LICENSE
```

I.Features & Functionality

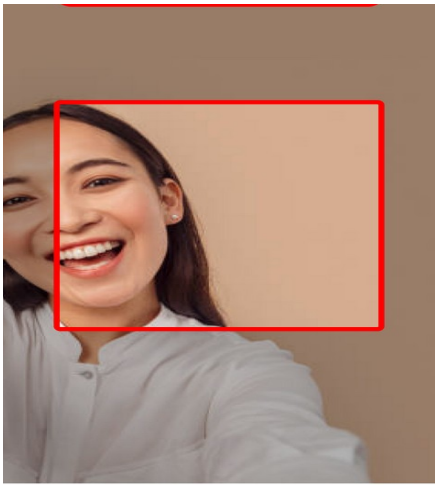
1. Permissions & Terms Flow

- Initial landing page requests:
    - Camera access (mandatory)
    - Optional microphone access
    - Concent to Terms of Service and Privacy Policy (non-orthogonal)

- Users cannot proceed without granting permissions.

2. Facial Recognition Login

- Detects faces using webcam.
- AI predicts:
    - Identity
    - Age
    - Gender
    - Ethnicity
- Liveness detection ensures the user is real (blink, gestures).
- Multi-factor authentication fallback (password/OTP).



The snap above is just a rough picture in mind.

3. Unauthorized Detection & Logging

- Unknown or mismatched faces:
    - Snapshot capture
    - Logs: image, timestamp, device info, AI estimates
    - Optional blockchain logging for immutable audit trail

4. Dashboard

- Real-time feed with:
    - Authorized vs unauthorized attempts
    - AI demographic overlays
    - IoT simulation (digital locks, LEDs)
- Analytics dashboard: attempt trends, demographics summary :

5. AI Analytics

- Predictive threat analysis
- Detection of unusual login patterns
- Optional admin insights for monitoring high-risk activity
- provide a summary of all the activitiesand can be downloadable as pdf or copied to clipboard [download summary button]

6. Advanced Features :

- Multi-camera support for enterprise environments
- NLP-based phishing detection
- Dark-web monitoring for stolen credentials
- Blockchain for compliance and immutable logs
- Predictive threat scoring using AI
- Hacker news section that constantly fetches hacker and cybersecurity-related news and emergents

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II.High-Performance Architecture (Blueprint)

Client-Side (Browser)

- React + face-api.js / TensorFlow.js
- WebRTC / getUserMedia for live webcam feed
- Lightweight detection, cropping, and sending embeddings to backend
- Displays live overlay, IoT simulation, and analytics

Backend (C++)

- Handles performance-critical AI tasks:
    - Face recognition & embedding extraction
    - Age, gender, ethnicity estimation
    - Liveness detection
- REST API / WebSocket communication with frontend
- Logs unauthorized access and triggers IoT simulation
- GPU acceleration via TensorRT or CUDA for real-time inference

*Tor-Enhanced Security:

- Backend runs with a Tor daemon
- Exposes a private `.onion` Hidden Service for secure admin access and log retrieval
- Supports Tor SOCKS5 proxy for outbound threat intelligence lookups
- Optional Tor identity rotation using ControlPort for OSINT functions
- Logs unauthorized access and triggers IoT simulation
- GPU acceleration via TensorRT or CUDA

Database

- PostgreSQL / MongoDB stores:
    - User embeddings
    - Event logs: timestamp, AI estimates, snapshots
- Optional blockchain stores immutable log hashes

IoT / Physical Layer

- Simulated digital locks, LEDs, buzzers
- Optional Raspberry Pi / Arduino for demonstration
- Authorized face → green LED/unlock, unauthorized → red LED/alert

Tor-Protected Camera Network (optional):

- External or remote cameras can expose video feeds as private `.onion` services
- The backend connects anonymously to these hidden camera nodes
- Simulated digital locks, LEDs, buzzers
- Optional Raspberry Pi / Arduino for demonstration
- Authorized face → green LED/unlock, unauthorized → red LED/alert

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III.Data Flow Diagram [DFD]:

```
1. User grants permissions → landing page
2. Webcam feed captured → lightweight detection in browser
3. Cropped face → C++ backend
4. AI predicts identity & demographics
5. Backend returns result → dashboard updates
6. Unauthorized face → snapshot logged, blockchain hash optional
7. IoT simulation triggered in dashboard / physical device
```

***DFD[TOR]:**

Free Hosting (Vercel/Netlify)

```
    -------------------------------
    |  React Dashboard UI        |
    |  Login Page + WebCam       |
    |  JS Face Detection          |
    -------------------------------
                |
                | fetch() requests
                v
  +-------------------------------------+
  |     Your Backend Machine (Linux)     |
  |  Running Tor + C++ AI Backend        |
  |  Hidden Service Enabled              |
  |  .onion address generated            |
  +-------------------------------------+
```

IV.Implementation Roadmap

Phase 1: Project Setup

- Folder structure: backend_cpp, frontend, database, docs, scripts
- Environment setup: OpenCV, dlib, TensorRT, Node.js, React, PostgreSQL/MongoDB

Phase 2: AI Model Preparation

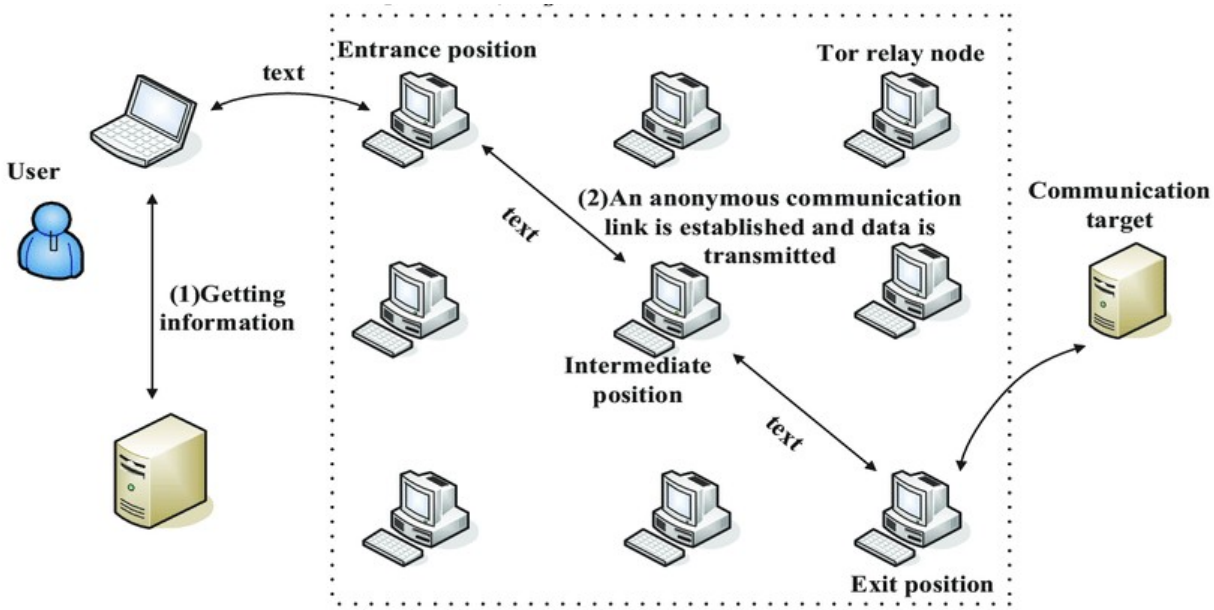
- FaceNet / Dlib embeddings for identity
- Age, gender, ethnicity models optimized for real-time inference
- Liveness detection implementation

Phase 3: C++ Backend Implementation

- Core AI engine for inference
- Logging & blockchain integration
- IoT triggers simulation

Tor Integration Tasks:

- Install and configure Tor daemon
- Enable Hidden Service for secure admin dashboard
- Add SOCKS5 proxy support for outbound intelligence queries
- Add ControlPort support for identity rotation
- Configure backend endpoints for `.onion` access



Phase 4: Frontend Dashboard

- Permissions & login pages
- Live webcam feed with overlays
- Real-time dashboard with logs & analytics
- Result: a clean modern-looking dashoard

***Dashboard looks && aesthetics wish [UI]:**

- Dark background with golden color features
- An accessability button drawer with the following features :
    - Dark mode and light mode switching – darkmode(pitch black) lightmode(deep green)
    - Language changer section that fully changes the interface language in real time [English-default, French, Swahili, German and Spanish]
    - An autoscroll button that has a silider to increase or decrease speed

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- An AI helper button (has an area for api pasting in the code)
- Random background dynamic doodles
- Fully glassy animated buttons and 3D movement while scrolling
- Modern looking graphs analysis and charts

Phase 5: Database & Storage

- Store embeddings, event logs, snapshots
- Encrypt sensitive data

Phase 6: Demo / Hackerthon Presentation

- Laptop webcam + dashboard + optional LED/buzzer
- Preloaded test faces (authorized vs unauthorized)
- Show AI recognition, logs, IoT response in real-time

Phase 7: Advanced Features

- Multi-camera support
- Phishing detection module
- Blockchain audit trail
- Predictive AI threat scoring

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